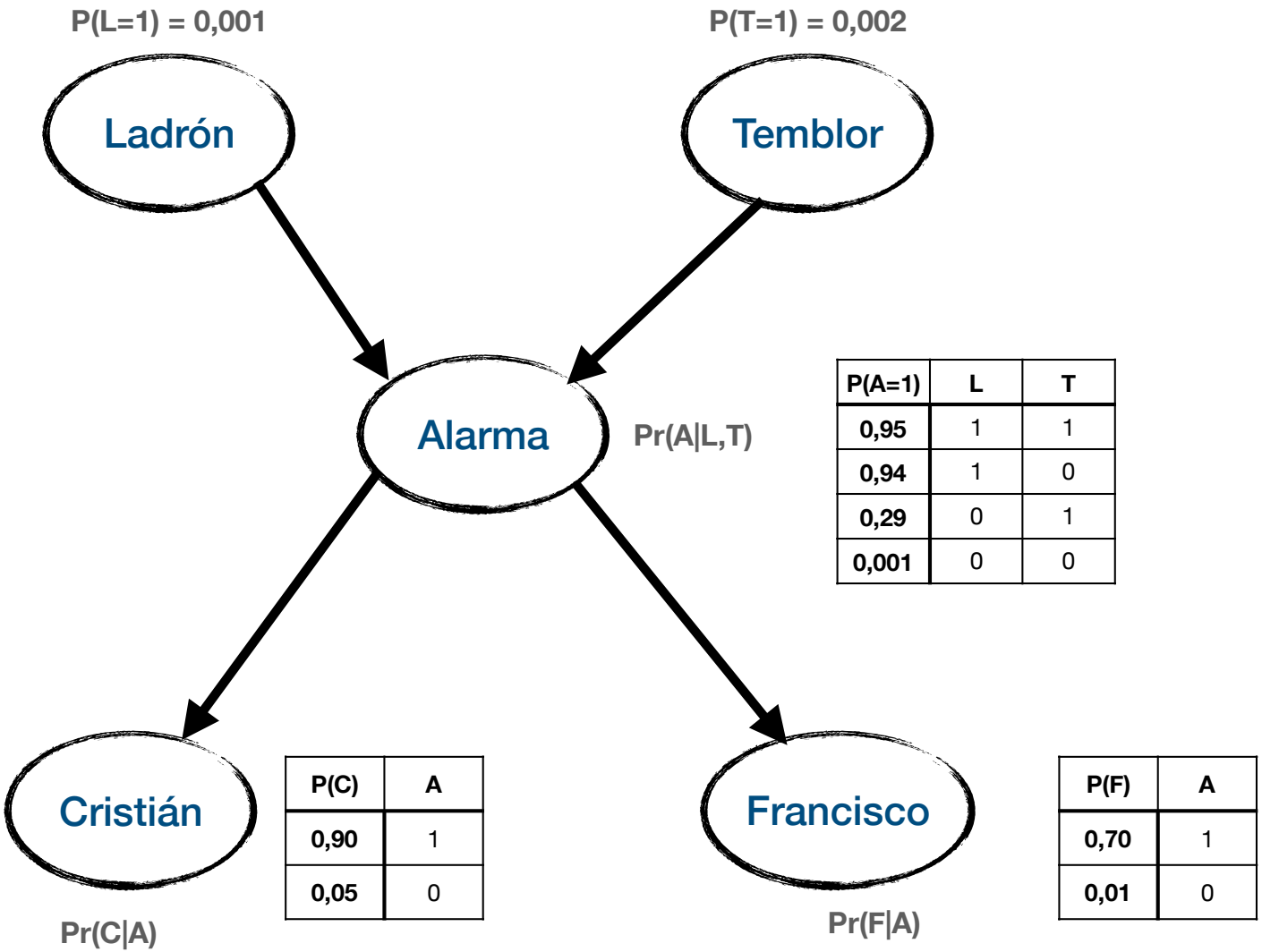
Razonamiento Probabilista

Repaso Probabilidades

Dr. Mauricio Araya

Redes Bayesianas

Inferencia



P(A=1)	L	T
0,95	1	1
0,94	1	0
0,29	0	1
0,001	0	0

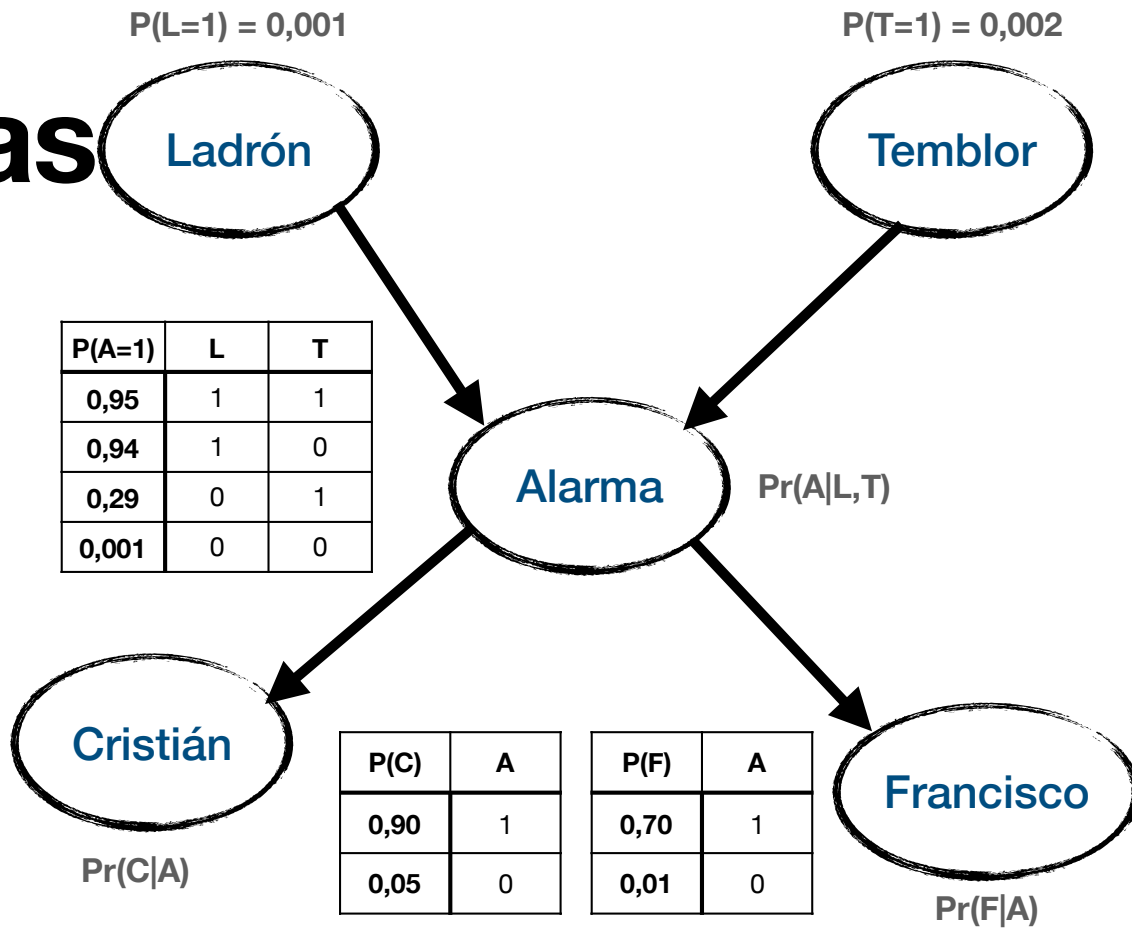
P(C)	A
0,90	1
0,05	0

P(F)	A
0,70	1
0,01	0

Redes Bayesianas

Inferencia

P(L=1)	C	F
?	1	1
?	1	0
?	0	1
?	0	0



P(A=1)	L	T
0,95	1	1
0,94	1	0
0,29	0	1
0,001	0	0

P(C)	A
0,90	1
0,05	0

P(F)	A
0,70	1
0,01	0

$$P(L = 1 \mid C = c, F = f) = \frac{P(C = c, F = f \mid L = 1)P(L = 1)}{P(C = c, F = f)}$$

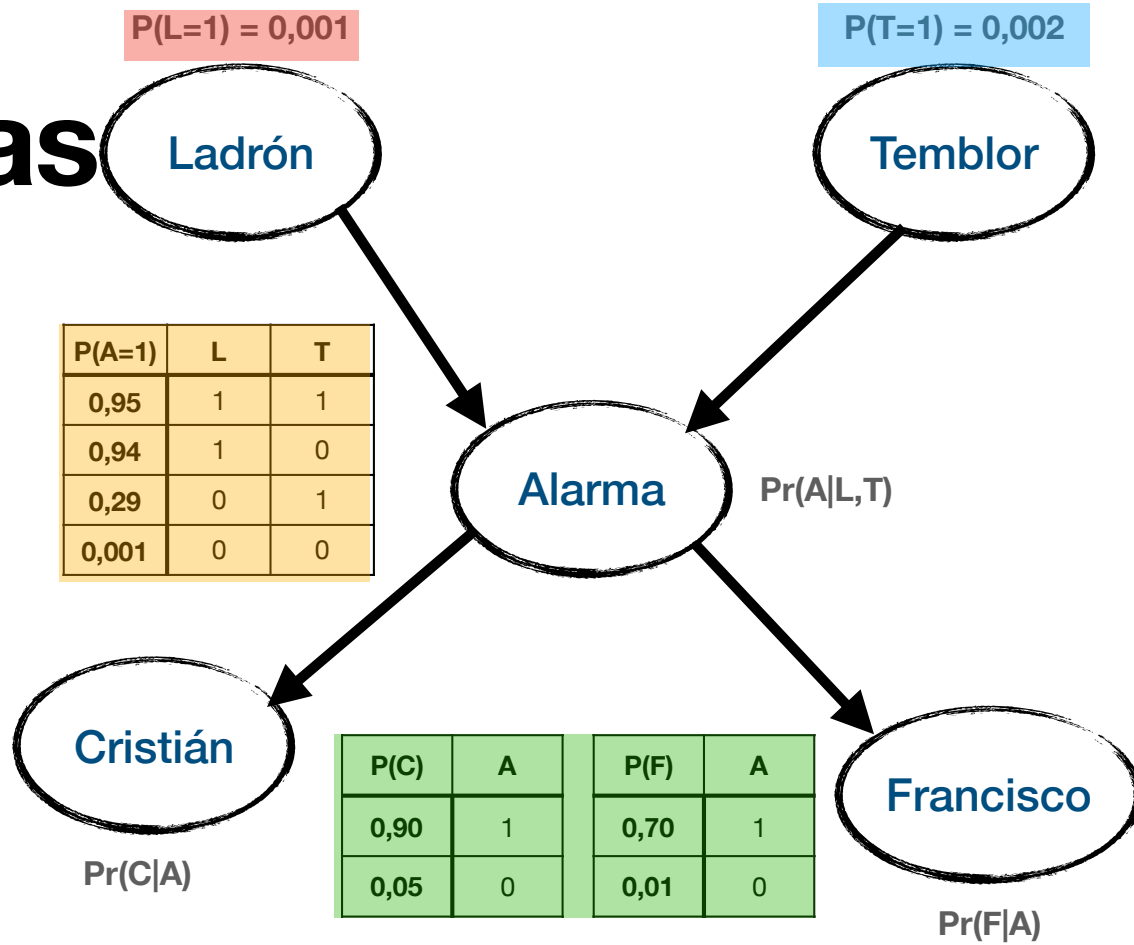
$$P(C = c, F = f \mid L = 1) = P(C = c, F = f \mid A = 1)P(A = 1 \mid L = 1) + P(C = c, F = f \mid A = 0)P(A = 0 \mid L = 1)$$

$$P(A = a \mid L = 1) = P(A = a \mid L = 1, T = 1)P(T = 1) + P(A = a \mid L = 1, T = 0)P(T = 0).$$

Redes Bayesianas

Inferencia

P(L=1)	C	F
?	1	1
?	1	0
?	0	1
?	0	0



P(A=1)	L	T
0,95	1	1
0,94	1	0
0,29	0	1
0,001	0	0

P(C)	A
0,90	1
0,05	0

P(F)	A
0,70	1
0,01	0

$$P(L = 1 \mid C = c, F = f) \propto P(C = c, F = f \mid L = 1) P(L = 1)$$

$$P(C = c, F = f \mid L = 1) = P(C = c, F = f \mid A = 1) P(A = 1 \mid L = 1) + P(C = c, F = f \mid A = 0) P(A = 0 \mid L = 1)$$

$$P(A = a \mid L = 1) = P(A = a \mid L = 1, T = 1) P(T = 1) + P(A = a \mid L = 1, T = 0) P(T = 0).$$

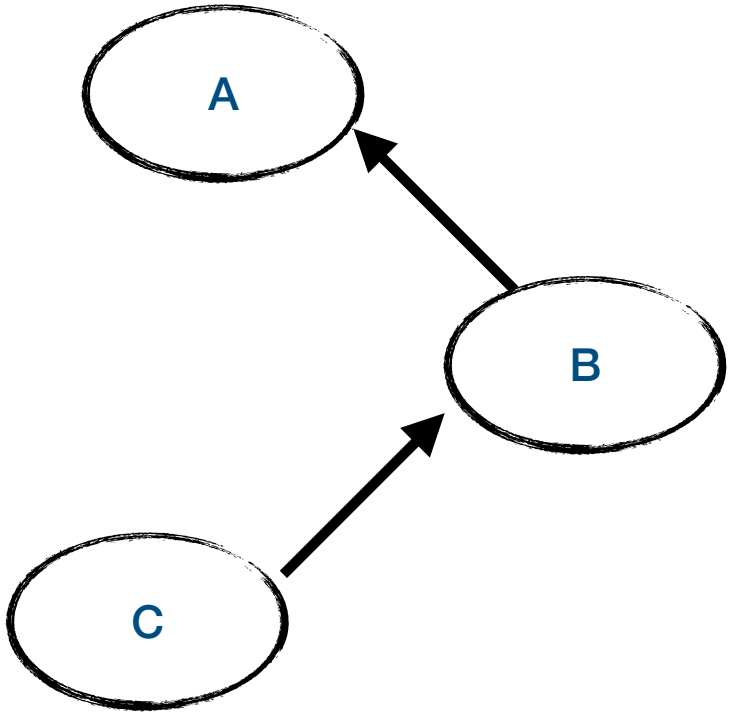
Razonamiento Probabilista

Espacios Grandes o Continuos!

$$P(A = a) = \int_b P(A = a|B = b)P(B = b)db$$

- Integración numérica
- Muestreo
- Simulación de Monte-Carlo

¿Mundos abiertos?



Razonamiento Probabilista

Variabilidad Aleatoria

$$y = g(x, z, \alpha, \beta, \gamma, \epsilon, \cdots, \omega)$$

$$y = f(x, z) + h(x, z, \alpha, \beta, \gamma, \epsilon, \cdots, \omega)$$

Función determinista (i.e. cinemática)

Función determinista (i.e. roce, relatividad, mecánica, sensores, etc)

$¿\alpha, \beta, \gamma, \epsilon, \cdots, \omega?$

$$y = f(x, z) + \varepsilon(x, z)$$

Función Estocástica

$$\varepsilon \sim Pr(\mathcal{E} = \varepsilon; \theta)$$

Distribución de Probabilidad

Razonamiento Probabilista

Variabilidad Aleatoria

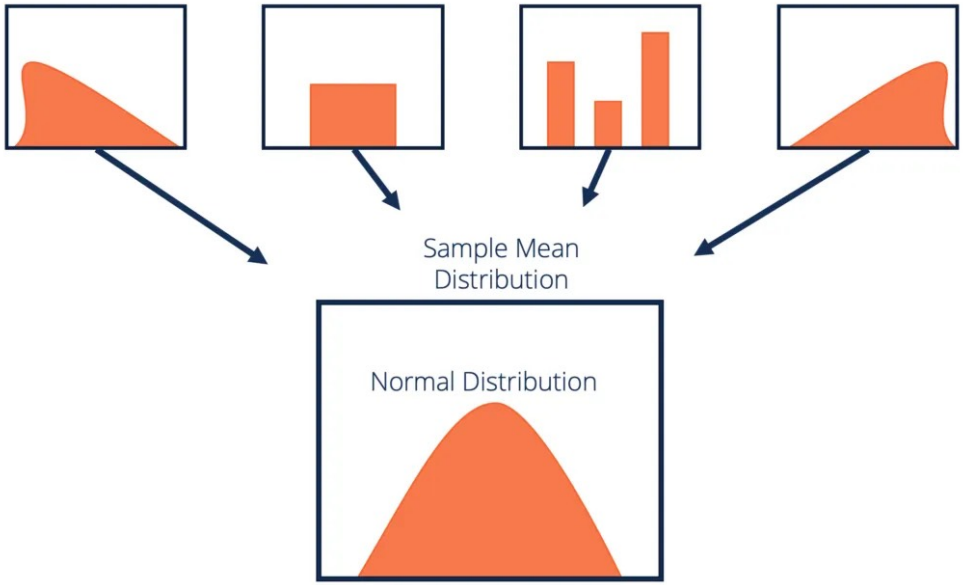
$$h(x, z, \alpha, \beta, \gamma, \epsilon, \cdots, \omega)$$

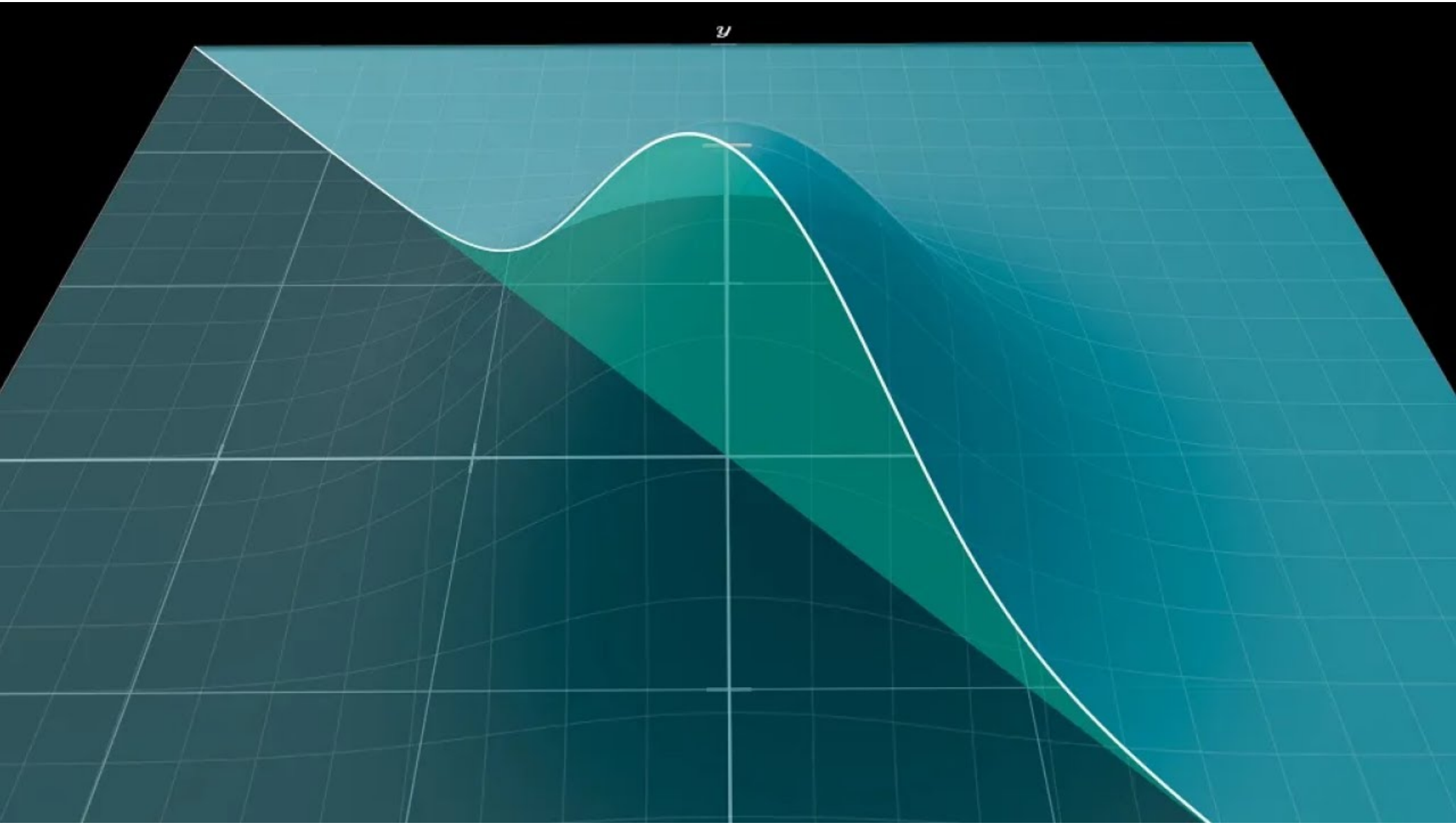
Función determinista

$$\varepsilon \sim Pr(\mathcal{E} = \varepsilon; \theta)$$

Distribución de Probabilidad






y

Razonamiento Probabilista

Variabilidad Aleatoria

Modela lo que no modelamos!
Inferir desde datos ruidosos
Base de Inferencia Estadística

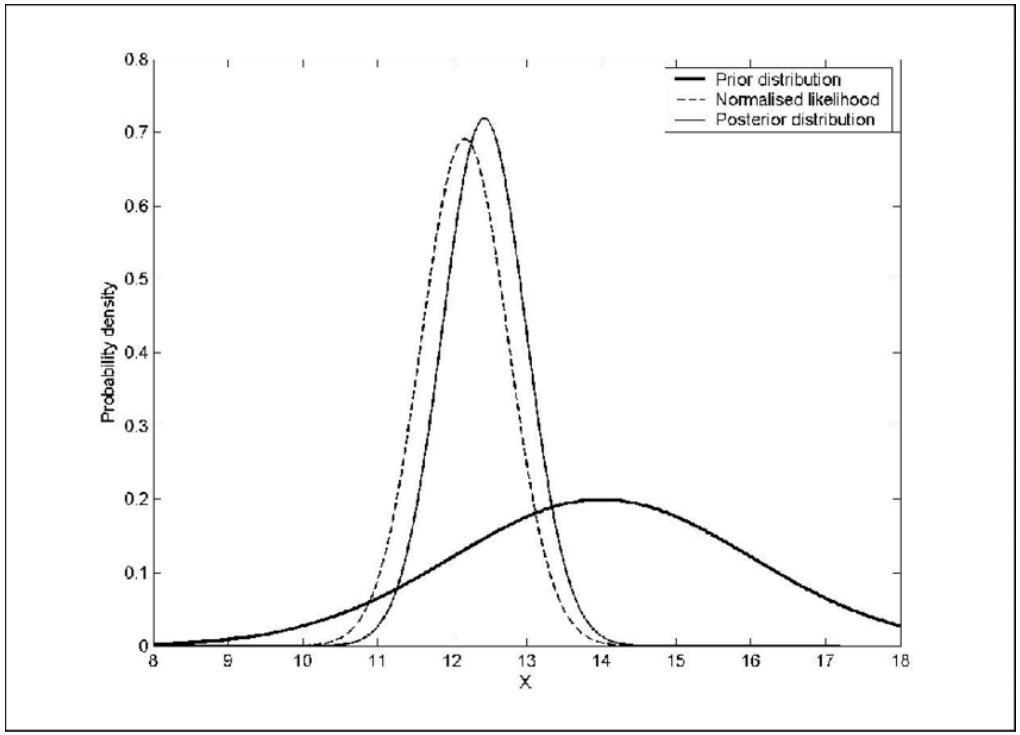
Razonamiento Probabilista

Incertidumbre Epistémica

$$y = f(x, z)$$

Función determinista

x es desconocido!



$$Pr(X = x|Y = y, z) = \frac{Pr(Y = y|X = x, z)Pr(X = x, z)}{Pr(Y = y|z)}$$

$$= \frac{f(x, z)Pr(X = x|z)}{\int_{x'} f(x', z)dP(x')}$$

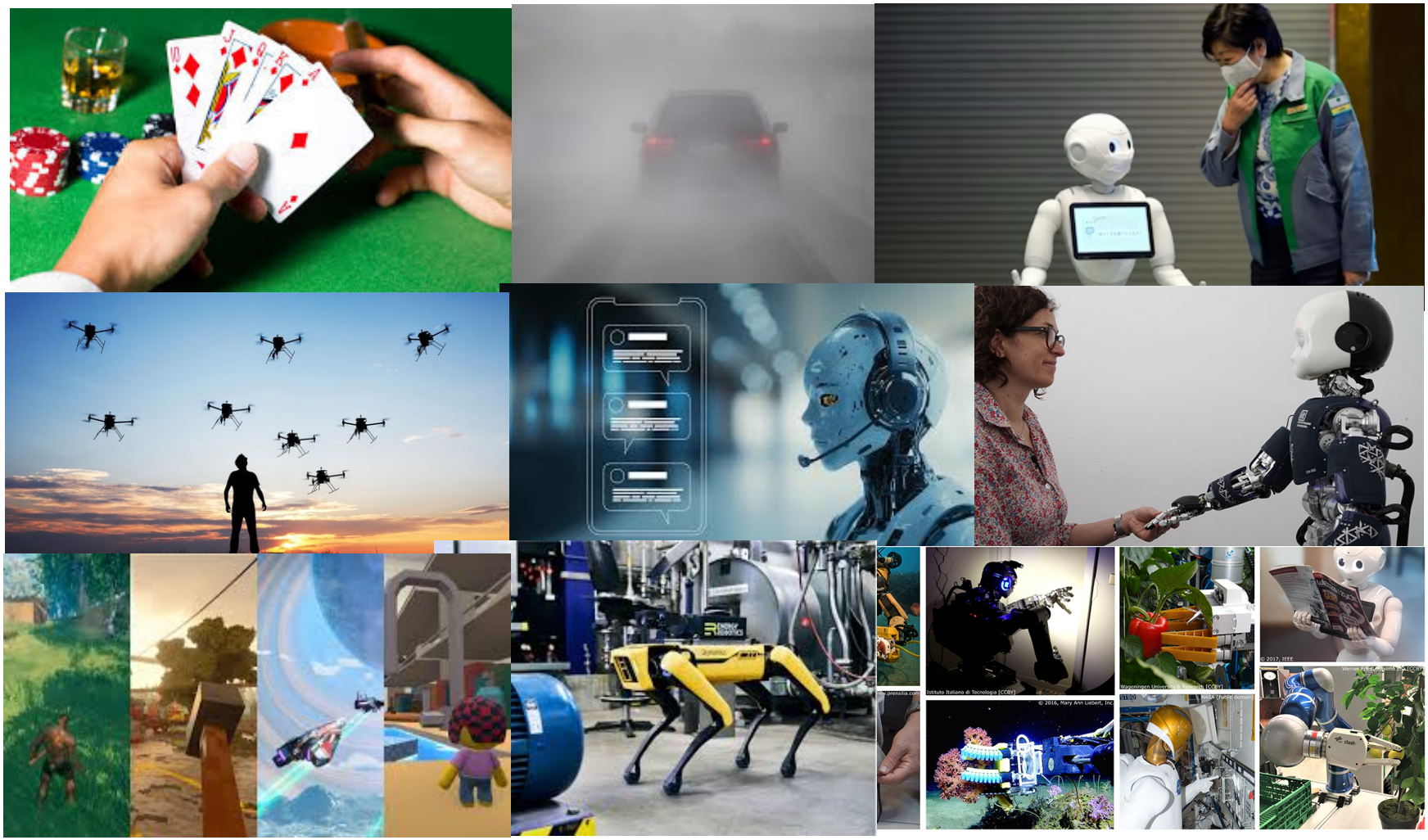
Razonamiento Probabilista

Incertidumbre Epistémica

Modela el **conocimiento** de variables
Base de la inferencia Bayesiana
Menos común que V.A.

Incomplete information

Everywhere…



Razonamiento Probabilista

Agentes Inteligentes

Dr. Mauricio Araya